A HENSELIAN LOCAL RING IS A FILTERED COLIMIT OF HENSELIAN LOCAL G-RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finitely generated algebras over the integers are noetherian local G-rings after localisation, by the ubiquity of G-rings. Proposition \ref{proposition-ubiquity-G-ring}, cited at proof lines 12684-12685. Makes the approximating rings G-rings.
- The henselization of a filtered colimit is the filtered colimit of the henselizations. Lemma \ref{lemma-henselization-colimit}, cited at proof lines 12686-12687. Replaces the colimit of the rings by the colimit of their henselizations.
- The henselization of a G-ring is a G-ring. Lemma \ref{lemma-henselization-G-ring}, cited at proof lines 12688-12689. Keeps the G-ring property along the colimit.

Source reference. Original lemma 0A41: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12759–12759. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07PX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9812–9812. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0A04: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12622–12622. Navigation pointer to a statement the

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Source reference. Original lemma 07QR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9841–9841. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a henselian local ring. Then A is a filtered colimit of a system of henselian local G-rings with local transition maps.*

Proof. Write $A = \operatorname{colim} A_i$ as a filtered colimit of finite type $\mathbf{Z}$-algebras. Let $\mathfrak{p}_i$ be the prime ideal of A_i lying under $\mathfrak{m}$. We may replace A_i by the localization of A_i at $\mathfrak{p}_i$. Then A_i is a Noetherian local G-ring (Proposition 07PX). By Lemma 0A04 we see that $A = \operatorname{colim} A_i^h$. By Lemma 07QR the rings A_i^h are G-rings. □